

LIPIDS

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INTRODUCTION

Importance

Hyperlipidaemia or high blood cholesterol is a major risk factor for coronary artery disease (CAD). The relationship between CAD and total cholesterol levels is continuous and curvilinear. Most of the cholesterol in the blood is carried on low density lipoprotein (LDL) particles, and elevated LDL cholesterol (LDL-C) causes atherosclerosis.

Low levels of high-density lipoprotein (HDL) cholesterol are associated with an increased risk of CAD and high levels with a decreased risk. Moderate to severe levels of hypertriglyceridemia progressively increase the risk of pancreatitis. Though also associated with CAD, this relationship is weakened when adjusted for other related variables.

Adopting a healthy lifestyle which includes good diet, sufficient physical activity and tobacco abstinence together with pharmacotherapy where indicated improves an individual's cardiovascular risk.

Epidemiology

In the National Population Health Survey 2022, about three in 10 Singapore residents aged 18 to 74 years reported that they had hyperlipidaemia during the period 2021-2022.

Males (36.2%) had higher prevalence of high blood cholesterol than females (27.9%). The prevalence of high blood cholesterol increased with age; from around one in 10 (9.6%) adults in the 18 to 29 years age group to more than one in two in the 60 to 74 years age group (60 to 69 years: 53.7%; 70 to 74 years: 55.1%).

SCREENING AND DIAGNOSIS

Screening

Whom to Screen?

- All patients aged 40 years and above.
- All patients aged 18 years and above with risk factors for CAD.
- All patients with established coronary heart disease, cerebrovascular disease, peripheral vascular disease, hypertension and diabetes mellitus, impaired fasting glycaemia, or impaired glucose tolerance irrespective of age.
- Screening of all first-degree relatives of diagnosed familial hypercholesterolemia patients is recommended (see section on [Familial Hypercholesterolemia](#)).

Screening for lipids should be performed as part of a global cardiovascular / cardiometabolic risk assessment. Clinicians should consider screening for other risk factors such as blood pressure and blood glucose.

* - Risk factors for CAD include:

- Diabetes mellitus.
- Multiple CAD risk factors (e.g., tobacco use, hypertension, impaired fasting glycaemia, or impaired glucose tolerance).
- A family history of cardiovascular disease before age 50 years in male relatives or before age 60 years in female relatives.
- A family history suggestive of familial hypercholesterolemia.

Screening may be done using a fasting or non-fasting Lipid Panel. **In individuals with a higher risk of lipid disorders or a strong family history (e.g., familial hyperlipidaemia), the fasting lipid profile is preferred.**

Screening should be done annually, except for low-risk individuals with screening results within the LDL-C target levels and not on lipid modifying drug therapy when screening can be repeated at 3-yearly intervals.

Screening with Non-Fasting Lipid Panel

Clinicians should note that when interpreting non-fasting lipid profiles for screening post-meal TG levels can be slightly higher while LDL-C levels can be slightly lower compared to fasting levels.

Clinicians who use the non-fasting Lipid Panel need to be aware of possible:

1. Over-diagnosis of Hypertriglyceridemia: Individuals with raised TG levels in the non-fasted state, who would have otherwise had normal TG levels in the fasted state.

Many of these individuals may still benefit from lifestyle intervention in the first instance. There may also be some cases of spuriously high TG readings that are secondary to very high fat intake prior to sampling. **Clinicians may consider repeating the fasting lipid profile to confirm the level and to guide treatment decisions**, especially when planning to start pharmacological therapy.

2. Under-diagnosis of Hyperlipidaemia: Individuals with normal/borderline LDL-C levels in the non-fasted state, who would otherwise have had elevated LDL-C levels in the fasted state.

The magnitude of the difference in LDL-C values in these scenarios is small and may not significantly alter the risk of cardiovascular disease over a 3-year period. Furthermore, these individuals should undergo repeat screening in the next 3 years, and there would be opportunities to identify them at subsequent testing if LDL-C levels remain elevated. **Depending on clinical judgment patients with results close to the threshold (based on their risk category) may need to be reviewed earlier with a fasting lipid profile.**

Possible scenarios for non-fasting Lipid Panel include low risk patients who request for screening with same day labs or patients deemed more suitable for non-fasting DM panel such as:

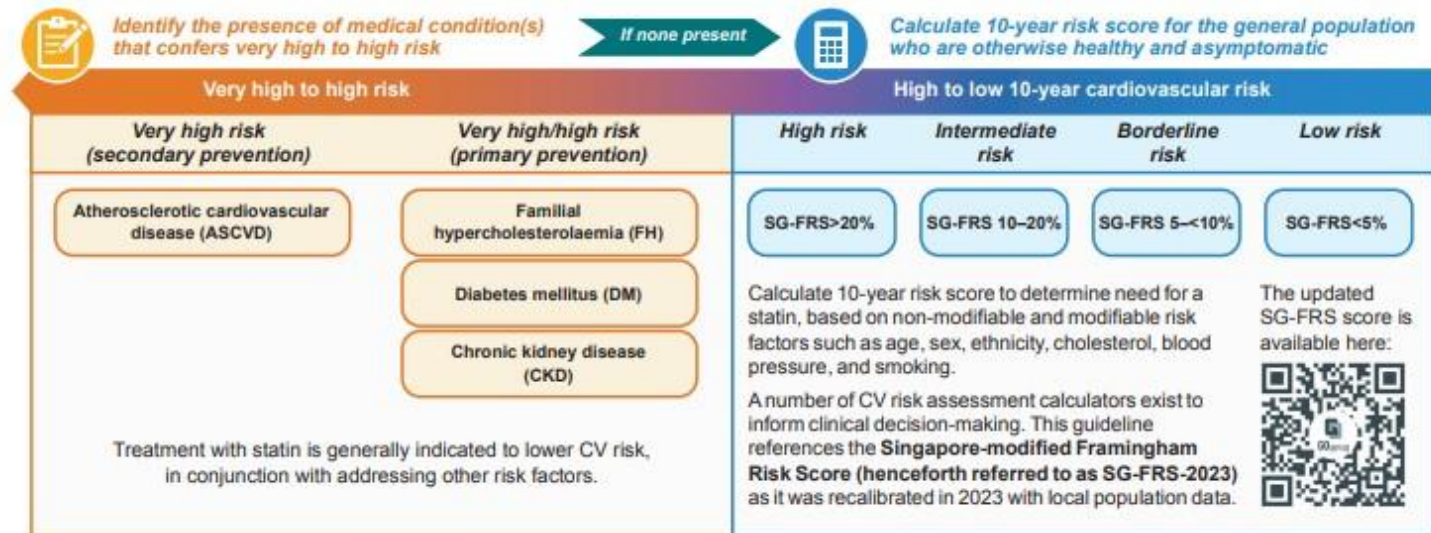
- No history of hypertriglyceridemia
- At risk for hypoglycaemia (frail, elderly, high dose insulin / sulphonylureas, chronic kidney disease)

Doctors who prefer to use fasting lipid tests for the initial screening of lipid disorders can continue to do so.

Risk Stratification and Treatment Goals

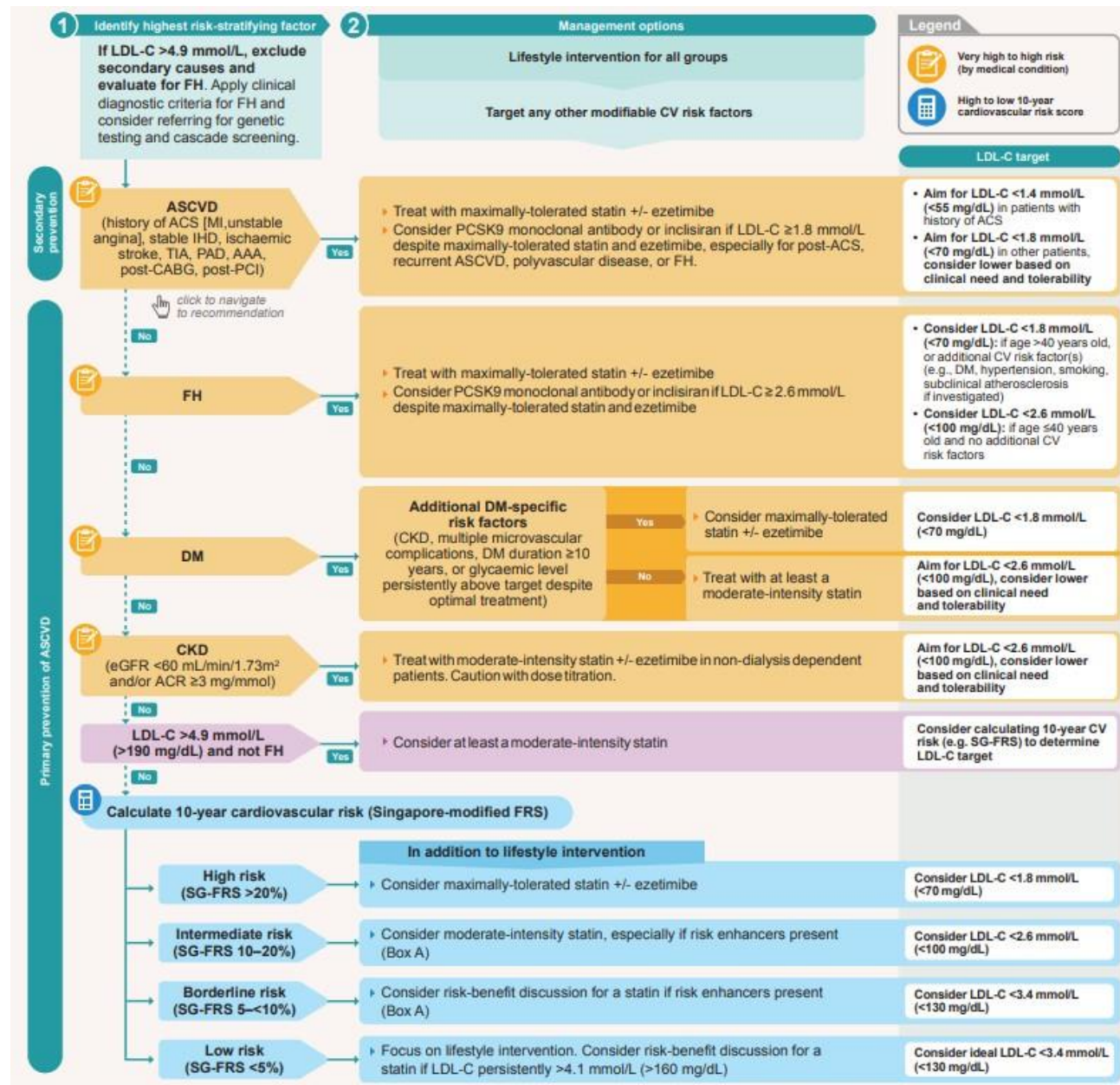
The overall cardiovascular (CV) risk provides the starting point for clinical judgment and shared decision-making to manage hyperlipidaemia, since the benefit from lipid lowering is proportionate to the baseline CV risk.

To establish the CV risk, assess for presence of medical conditions that confer risk and other CV risk factors. In the absence of conditions that confer very high / high level of risk, calculate the 10-year risk score using SG-FRS-2023.



Select a statin of appropriate intensity for patients with conditions that confer very high / high risk. Consider further intensification of lipid-lowering therapy if LDL-C is still above target despite appropriate intensity statin.

For patients without established conditions, consider statin therapy based on 10-year risk.



(From Lipid management: Focus on cardiovascular risk. Agency for Effective Care, MOH)

Singapore-Modified Framingham Risk Score 2023 (SG-FRS-2023) (Men)

Table 1.1 Estimation of 10-Year Coronary Artery Disease Risk for *Men*

Age	Points
20 - 34	-9
35 - 39	-4
40 - 44	0
45 - 49	3
50 - 59	6
55 - 59	8
60 - 64	10
65 - 69	11
70 - 74	12
75 - 79	13

HDL Cholesterol mmol/L (mg/dL)	Points
≥ 1.6 (60)	-1
1.3 - 1.5 (50 - 59)	0
1.0 - 1.2 (40 - 49)	1
< 1.0 (40)	2

Total Cholesterol mmol/L (mg/dL)	Points				
	Age 20 - 39	Age 40 - 49	Age 50 - 59	Age 60 - 69	Age 70 - 79
< 4.1 (160)	0	0	0	0	0
4.1 - 5.1 (160 - 199)	4	3	2	1	0
5.2 - 6.1 (200 - 239)	7	5	3	1	0
6.2 - 7.2 (240 - 279)	9	6	4	2	1
≥ 7.3 (280)	11	8	5	3	1

Systolic BP (mmHg)	Points	
	If untreated	If treated
< 120	0	0
120 - 129	0	1
130 - 139	1	2
140 - 159	1	2
≥ 160	2	3

Smoker	Points				
	Age 20 - 39	Age 40 - 49	Age 50 - 59	Age 60 - 69	Age 70 - 79
No	0	0	0	0	0
Yes	8	5	3	1	1

1. Estimate the individual's 10-year CAD risk by allocating points based on his age, total and HDL cholesterol levels, smoking status, and systolic blood pressure (BP).
2. Check the total points against Table 1.2 to estimate that individual's 10-year CAD risk.

Table 1.2 Estimation of 10-Year Coronary Artery Disease Risk for *Men*

Total Points	10 Year Risk (%)		
	Chinese	Malay	Indian
-5	< 1	< 1	< 1
-4	< 1	< 1	< 1
-3	< 1	< 1	< 1
-2	< 1	< 1	< 1
-1	< 1	< 1	< 1
0	< 1	< 1	< 1
1	< 1	< 1	1
2	< 1	1	1
3	< 1	1	1
4	1	1	1
5	1	1	2
6	1	2	2
7	1	2	3
8	2	3	4
9	2	3	5
10	3	4	6
11	3	5	7
12	4	7	10
13	5	9	12
14	7	11	15
15	9	14	19
16	11	18	24
17	14	22	30
18	18	28	37
19	23	34	45
20	28	42	54

10-Year Coronary Artery Disease Risk (Women)

Table 2.1 Estimation of 10-Year Coronary Artery Disease Risk for Women

Age	Points
20 - 34	-7
35 - 39	-3
40 - 44	0
45 - 49	3
50 - 59	6
55 - 59	8
60 - 64	10
65 - 69	12
70 - 74	14
75 - 79	16

HDL Cholesterol mmol/L (mg/dL)	Points
≥ 1.6 (60)	-1
1.3 - 1.5 (50 - 59)	0
1.0 - 1.2 (40 - 49)	1
< 1.0 (40)	2

Total Cholesterol mmol/L (mg/dL)	Points				
	Age 20 - 39	Age 40 - 49	Age 50 - 59	Age 60 - 69	Age 70 - 79
< 4.1 (160)	0	0	0	0	0
4.1 - 5.1 (160 - 199)	4	3	2	1	0
5.2 - 6.1 (200 - 239)	8	6	4	2	1
6.2 - 7.2 (240 - 279)	11	8	5	3	2
≥ 7.3 (280)	13	10	7	4	2

Systolic BP (mmHg)	Points	
	If untreated	If treated
< 120	0	0
120 - 129	1	3
130 - 139	2	4
140 - 159	3	5
≥ 160	4	6

Smoker	Points				
	Age 20 - 39	Age 40 - 49	Age 50 - 59	Age 60 - 69	Age 70 - 79
No	0	0	0	0	0
Yes	9	7	4	2	1

1. Estimate the individual's 10-year CAD risk by allocating points based on his age, total and HDL cholesterol levels, smoking status, and systolic blood pressure (BP).
2. Check the total points against Table 2.2 to estimate that individual's 10-year CAD risk.

Table 2.2 Estimation of 10-Year Coronary Artery Disease Risk for Women

Total Points	10 Year Risk (%)		
	Chinese	Malay	Indian
0	< 1	< 1	< 1
1	< 1	< 1	< 1
2	< 1	< 1	< 1
3	< 1	< 1	< 1
4	< 1	< 1	< 1
5	< 1	< 1	< 1
6	< 1	< 1	< 1
7	< 1	< 1	< 1
8	< 1	< 1	< 1
9	< 1	< 1	1
10	< 1	1	1
11	< 1	1	1
12	< 1	1	1
13	1	1	2
14	1	1	2
15	1	2	3
16	1	2	3
17	2	3	4
18	2	4	6
19	3	5	7
20	4	7	10
21	5	9	12
22	7	11	16
23	8	14	20
24	11	18	25
25	14	23	31
26	18	29	39
27	22	36	47

Secondary Dyslipidaemia

Secondary dyslipidaemia may occur in the various conditions listed below. These conditions should be excluded in any patient presenting with dyslipidaemia.

Common Causes of Secondary Dyslipidaemia

Causing Increased Total Cholesterol and LDL-C

Hypothyroidism

Nephrosis

Cholestatic diseases of the liver due to abnormal lipoproteins, e.g., Primary Biliary Cirrhosis

Progestin or anabolic steroid treatment

Dysgammaglobulinemia (SLE, Multiple myeloma)

Protease inhibitors for treatment of HIV infection

Causing Increased Triglyceridemia and VLDL-C

Type 2 Diabetes

Chronic renal failure

Excessive Alcohol intake

Hypothyroidism

Obesity

Antihypertensive medications (thiazide diuretics and B-blockers)

Corticosteroid therapy (or severe stress that increases endogenous corticosteroid)

Orally administered oestrogens, OCPs, pregnancy

Protease inhibitors for treatment of HIV infection

Hypertriglyceridemia

Triglyceride (TG) levels >1.7 mmol/L are considered elevated. Several studies suggest that elevated TG levels are associated with CAD. However, this association is significantly attenuated after adjustment for other blood lipid levels. There is a lack of clear evidence that TG reduction in itself improves CV risk.

Individuals with very high levels of TG, i.e., > 4.5 mmol/L or especially > 10 mmol/L, have an increased risk of acute pancreatitis and should be treated to reduce this risk. In these patients, the priority is to reduce the TG level to prevent acute pancreatitis.

Patients with moderately elevated TG not at risk of pancreatitis may also have concurrent CV risk. They should be risk stratified and treated with a statin if unable to achieve target LDL. Statins reduce TG levels by 10 – 20%.

For patients with TG ≤ 1.7mmol/L, LDL-M can be considered in place of lipid panel for monitoring. However, all patients with hyperlipidaemia should still have a full lipid panel (including TG, HDL etc.) done once a year as part of their annual fasting panel.

Familial Hypercholesterolemia (FH)

Patients with LDL >4.9 mmol/L should be considered for evaluation of FH. Patients can be referred to Genomic Assessment Centre (GAC) for genetic test after excluding secondary causes. For more information on NUP FH referral workflow, please click [here](#).

Patients with FH are individuals at high / very high CV risk and goal LDL-C should be 2.6 or 1.8 mmol/L depending on additional CV risk (refer page 7). All first-degree relatives of patients with FH should be screened and screening for children can start from 2 years of age.

History-taking can be directed to assess for FH through clinical diagnostic criteria, such as the Dutch Lipid Clinic Network (DLCN) or Simon Broome Trust criteria. DLCN is recommended over Simon Broome criteria. Lifelong statin therapy in this population reduces the onset and progression of atherosclerosis.

Consider treating patients with DLCN ≥ 6 as clinical FH with or without a genetic test. A positive genetic test confirms the diagnosis of FH.

Table 3 - Dutch Lipid Clinic Network Criteria for Familial Hypercholesterolemia

Criterion	Score
Genetic Investigation	
Verified causal genetic mutation (LDLR; ApoB; PCSK9)	8
Biochemical Investigation	
LDL-C ≥ 8.5 mmol/L	8
LDL-C 6.5 – 8.4 mmol/L	5
LDL-C 5.0 – 6.4 mmol/L	3
LDL-C 4.0 – 4.9 mmol/L	1
Clinical Investigation	
Tendon Xanthomata	6
Arcus Cornealis (age < 45 years old)	4
Family Medical History	
1 st degree relative with *premature coronary artery disease OR 1 st degree relative with verified LDL-C above 95 th percentile	1
1 st degree relative with tendon xanthomata or arcus cornealis OR Children with verified LDL-C above the 95 th percentile	2
Patient Past Medical History	
*Premature coronary artery disease	2
*Premature cerebral or peripheral artery disease	1
Diagnosis	
Definite FH – points	≥ 8
Probable FH – points	6 – 7
Possible FH – points	3 – 5
Unlikely FH – points	< 3

**Premature disease is defined as disease before the age of 55 years (men) or 60 years (women)*

Simon Broome Trust Diagnostic Criteria for Familial Hypercholesterolemia

1. Diagnose a person with **Definite** Familial Hypercholesterolemia if they have
 - Cholesterol concentrations as defined in Table 4 **and** tendinous xanthomata in the patient or a first degree relative; **OR**
 - DNA-based evidence of an LDL-receptor mutation, familial defective apo B-100, or a PCSK9 mutation.
2. Diagnose a person with **Probable** Familial Hypercholesterolemia if they have cholesterol concentrations as defined in Table 4 **and at least one of the following:**
 - Family history of myocardial infarction: aged younger than 50 years in second-degree relative or aged younger than 60 years in first-degree relative.
 - Family history of raised total cholesterol greater than 7.5 mmol/L in adult first- or second-degree relative or greater than 6.7 mmol/L in child, brother or sister aged younger than 16 years.

Table 4 - Cholesterol levels to be used as diagnostic criteria for the index individual (Levels either pre-treatment or highest on treatment)

	Total Cholesterol	LDL-C
Child / young person aged under 16 years	> 6.7 mmol/L	> 4.0 mmol/L
Adult	> 7.5 mmol/L	> 4.9 mmol/L

MANAGEMENT

Goals of Treatment and Targets

- A basic principle in the prevention of CAD is that the intensity of risk reduction therapy should be adjusted to a person's risk of developing future coronary events. E.g., Start a patient on an appropriate intensity of statin to achieve the desired LDL goal efficiently (consider using the statin conversion table on page 18 to guide initiation / increment).
- Treat the whole patient and associated conditions / risk factors.
- For treatment targets, see section on [Risk Stratification and Treatment Goals](#).

Recommended Follow-Up Interval

Criteria	Recommended TCU Frequency	Recommended for alternate Dr / CM visit
• Achieved treatment target, AND • Does not have complication	6 - 12 months	√
• Achieved fair level of control related to individual target over the past 4 - 6 months	3 - 4 months	√
• Has not achieved treatment target, AND • Started on statin OR • Requires titration of medication	2 - 3 months	√

Lifestyle Modification and Patient Education

Appropriate lifestyle changes are an integral part of dyslipidaemia management. Lifestyle interventions can reduce risk of cardiovascular disease.

- Patients who smoke should be advised to stop smoking immediately. Consider referring to the pharmacy led smoking cessation clinic where nicotine replacement therapy (Medisave claimable) can be offered. Patients can also be directed to the "I Quit" programme by Health Promotion Board (HPB).
- Weight reduction of 5 – 10% in patients with body mass index above 23 kg/m² can improve their lipid profile – especially triglycerides.

- Persons with dyslipidaemia should undertake 150 to 300 minutes per week of moderate intensity physical activity spread out over 5 to 7 days per week. Exercise prescription should be individualised and appropriately paced especially in sedentary patients.
- A diet **rich** in wholegrains, vegetables, fruit, legumes, nuts, healthy sources of protein (e.g., fish, lean meat, tofu) and unsaturated oils and **low** in saturated and trans-fat, refined grains and cholesterol should be encouraged.
- For good overall health, individuals who do not currently drink alcohol should not start. For individuals who do drink alcohol, a maximum of two standard drink per day for women and three per day for men is recommended. Patients with elevated TG above 4.5 mmol/L should be encouraged to abstain from alcohol.

Drug Therapy / Medication

Drug	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions
HMG CoA Reductase Inhibitors ("Statin")				
Simvastatin (10 mg and 20 mg Tablets)	10 mg to 40 mg ON	40 mg/day	Headache, myalgia, fatigue, constipation, flatulence, dyspepsia, nausea, abdominal pain	Use with caution in patients with renal impairment, history of liver disease, elderly. Contraindicated in active liver disease; persistent elevations of serum transaminases > 3 times the upper limit of the normal (ULN) range; creatine kinase > 5x ULN; pregnancy; breastfeeding. In patients with moderate to severe hypertriglyceridemia initiated on fibrate therapy, if LDL-C remains elevated, a statin can be added to reduce cardiovascular risk. Patients already on simvastatin > 40 mg may be maintained on their dose only if they have been taking that dose for 12 or more months without evidence of muscle toxicity. In rare cases where Rosuvastatin at doses higher than 20 mg is indicated, initiation of therapy should be under close specialist supervision.
Atorvastatin (Lipitor®) (10mg, 20mg, 40mg and 80mg Tablets)	10 mg to 80 mg OD	80 mg/day		
Rosuvastatin (Crestor®) (10mg Tablet)	5 mg to 20 mg OD	40 mg/day		
Lovastatin (20 mg Tablet)	10 mg to 40 mg ON	80 mg/day		
Pravastatin (10 mg and 20 mg Tablet)	10 mg to 40 mg ON	80 mg/day		

Statin Conversion Table and Relative Cost[^]

% Effect on LDL-c	Simvastatin* (S1)		Atorvastatin (S2)		Rosuvastatin (S2)		Lovastatin (S1)		Pravastatin (S2)	
-27	10 mg	\$					20 mg	\$	20 mg	\$\$\$\$
-34	20 mg	\$	10 mg	\$\$	5 mg	\$	40 mg	\$	40 mg	\$\$\$\$\$
-41	40 mg	\$	20 mg	\$\$	10 mg	\$\$	80 mg	\$\$	80 mg	\$\$\$\$\$\$
-48			40 mg	\$\$	20 mg	\$\$				
-55			80 mg	\$\$\$	40 mg	\$\$\$				

Reference: Drug Information Handbook

* The recommended maximum dose of Simvastatin for patients on Amlodipine is 20 mg.

[^] Cost \$ <\$10 per month \$\$\$\$ \$30 - <\$60 per month
 \$\$ \$10 - <\$20 per month \$\$\$\$\$ \$60 - <\$90 per month
 \$\$\$ \$20 - <\$30 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Drug	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions
Cholesterol Absorption Inhibitor				
Ezetimibe (Ezetrol®) (10 mg tablet)	10 mg OD	10 mg OD	Arthralgia, dizziness, upper respiratory tract infection, diarrhoea, GGT increase. (With statin) AST or ALT increase, myalgia, fatigue, headache	For secondary prevention in patients with established CAD, ezetimibe, when added to a statin, produces further lowering of LDL-C and cardiovascular events. Liver function monitoring should be performed during coadministration. Use with caution in patients with severe renal impairment (creatinine clearance < 30 ml/min) or mild hepatic impairment (Child-Pugh class A). Not recommended for use with moderate or severe hepatic impairment (Child Pugh class B and C). Contraindication: Active hepatic disease or unexplained persistent elevations in serum transaminase.

Drug	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions
PCSK9 Inhibitor Monoclonal Antibody				
Evolocumab (prescription in NUP limited to patients with FH)	SC 140 mg once every 2 weeks	140 mg once every 2 weeks (fixed dose)	Injection site reactions, upper respiratory tract signs and symptoms, pruritus, arthralgia, back pain	Consider Evolocumab if LDL-C targets not reached despite maximally tolerated statin ± ezetimibe, especially for post-ACS, recurrent ASCVD, polyvascular disease, or FH. Route of administration and cost should be discussed before initiation of Evolocumab. Hepatic impairment: Limited experience in severe hepatic impairment (Child-Pugh class C). No dose adjustment is needed for patients with mild or moderate hepatic impairment. Renal impairment: No dosage adjustment is needed. eGFR < 30 ml/min: Unlikely to be significantly dialysed. No dosage adjustment is needed. Contraindication: Serious hypersensitivity to the PCSK9 inhibitor Apart from lipid panel, there is no need for routine monitoring of biochemical markers for patients on Evolocumab.

Drug	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions					
Fibric Acid Derivatives (“Fibrates”)									
Fenofibrate (100 mg and 300 mg capsule; Lipanthyl® 160 mg micronized tablet)	Capsule: 100 mg to 300 mg OD Micronized tablet: 160 mg OD	<u>Non-micronized formulation</u> - 400 mg/day in divided dose (Max 300 mg/day if used with a statin) <u>Micronized formulation</u> - Lipanthyl® Supra 160 mg/day (Max 200 mg/day if used with a statin)	Abdominal pain, nausea, vomiting, diarrhoea, flatulence, transaminases increased.	Fibrates can be used in patients with chronic kidney disease in stage 1 to 3, but the dosages should be reduced, with appropriate monitoring for side effects. When creatinine clearance is less than 30 ml/min, fibrates are contraindicated. Contraindicated in primary biliary cirrhosis, pre-existing gallbladder disease. Gemfibrozil should not be used in combination with statins. Gemfibrozil 300 mg capsule was delisted from NUP formulary in December 2024. The proposed alternative is fenofibrate and the dosing guidance is as follows:					
Gemfibrozil (Not available in NUP)	300 mg to 600 mg BD	1,200 mg/day		<table><tr><th>Scenario</th><th>Alternative</th></tr><tr><td>If Triglyceride level is well controlled with: • Gemfibrozil 600 mg BD or • Gemfibrozil 300 mg BD </td><td>Consider converting to: • Fenofibrate 100 mg OM </td></tr><tr><td>If Triglyceride level is not very well controlled with: • Gemfibrozil 600 mg BD </td><td>Consider converting to: • Fenofibrate 200 mg OM or • Lipanthyl Supra 160 mg </td></tr></table>	Scenario	Alternative	If Triglyceride level is well controlled with: • Gemfibrozil 600 mg BD or • Gemfibrozil 300 mg BD	Consider converting to: • Fenofibrate 100 mg OM	If Triglyceride level is not very well controlled with: • Gemfibrozil 600 mg BD
Scenario	Alternative								
If Triglyceride level is well controlled with: • Gemfibrozil 600 mg BD or • Gemfibrozil 300 mg BD	Consider converting to: • Fenofibrate 100 mg OM								
If Triglyceride level is not very well controlled with: • Gemfibrozil 600 mg BD	Consider converting to: • Fenofibrate 200 mg OM or • Lipanthyl Supra 160 mg								
Bile Acid Sequestrants									
Cholestyramine (Not available in NUP)	4 g once to four times daily	16 g/day	Constipation, nausea, vomiting, abdominal pain, headache	Contraindicated in patients with complete biliary obstruction. Drug interactions: To space at least 1 hour before or 4 hours after other medications.					

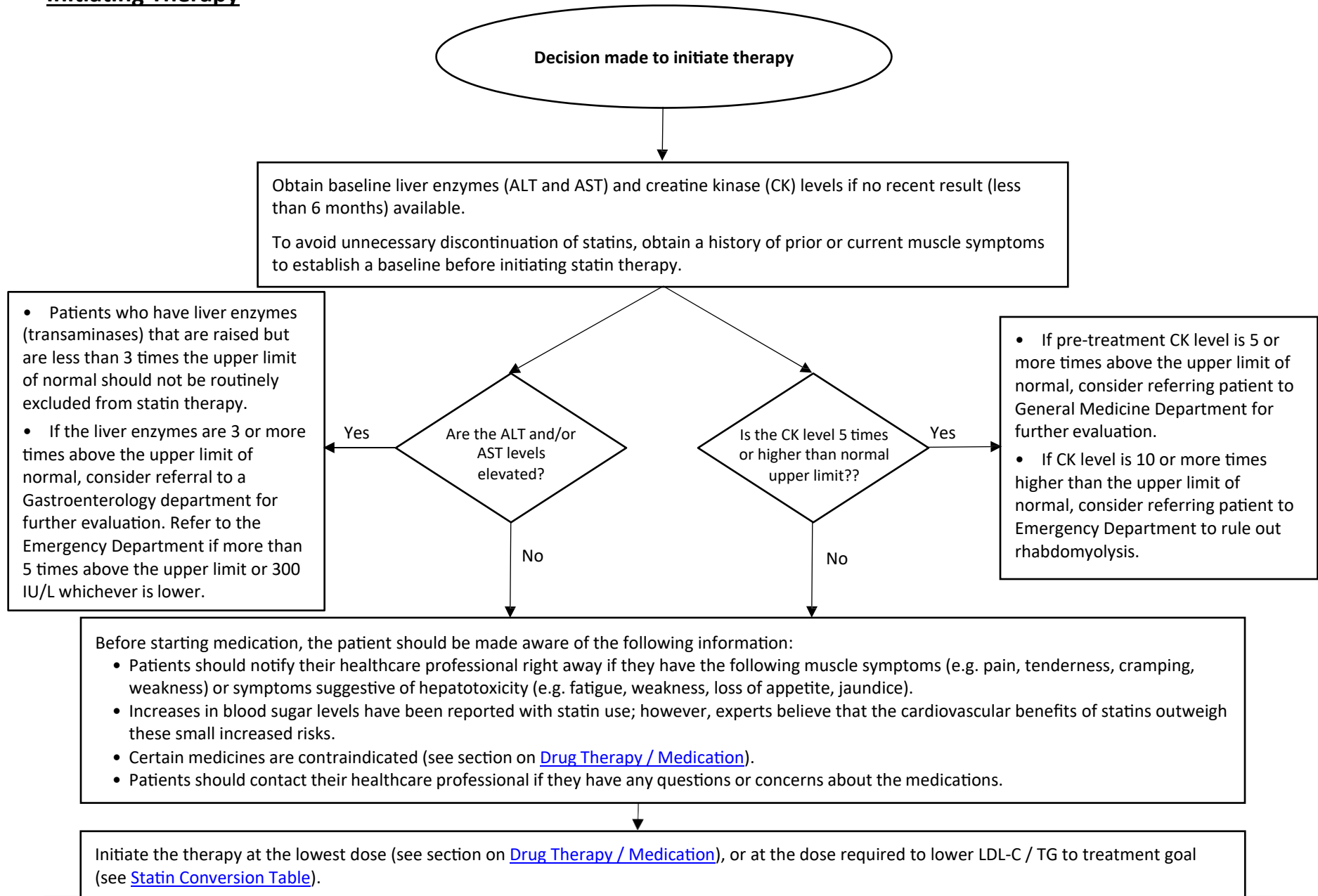
Drug	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions
Omega-3-Fish Acid Ethyl Esters				
Omega 3 Fish Oils (Available OTC)	2 g to 4 g/day EPA and DHA in single or 2 divided doses	4 g/day	Fishy aftertaste; Gastrointestinal side effects like abdominal bloating / pain, diarrhoea. May worsen LDL-C	In severe hypertriglyceridemia where fibrates alone may not adequately lower the markedly elevated TG levels, omega 3 fish oils should be added. Omega 3 fish oils have no effect on LDL cholesterol and cardiovascular mortality. Thus, they should not be used as a substitute for statins. Use with caution of high dose fish oil (> 3 g/day) in patients at risk of bleeding or high level of LDL-C.

Lovastatin Dose Limitations
Contraindicated with: • Itraconazole • Ketoconazole • Posaconazole • Erythromycin • Clarithromycin • Telithromycin • HIV protease inhibitors • Boceprevir • Telaprevir • Nefazodone
Avoid with lovastatin: • Cyclosporine • Gemfibrozil
Do not exceed 20 mg lovastatin daily with: • Danazol • Diltiazem • Verapamil
Do not exceed 40 mg lovastatin daily with: • Amiodarone • Ticagrelor
Avoid large quantities of grapefruit juice (> 1 litre daily)

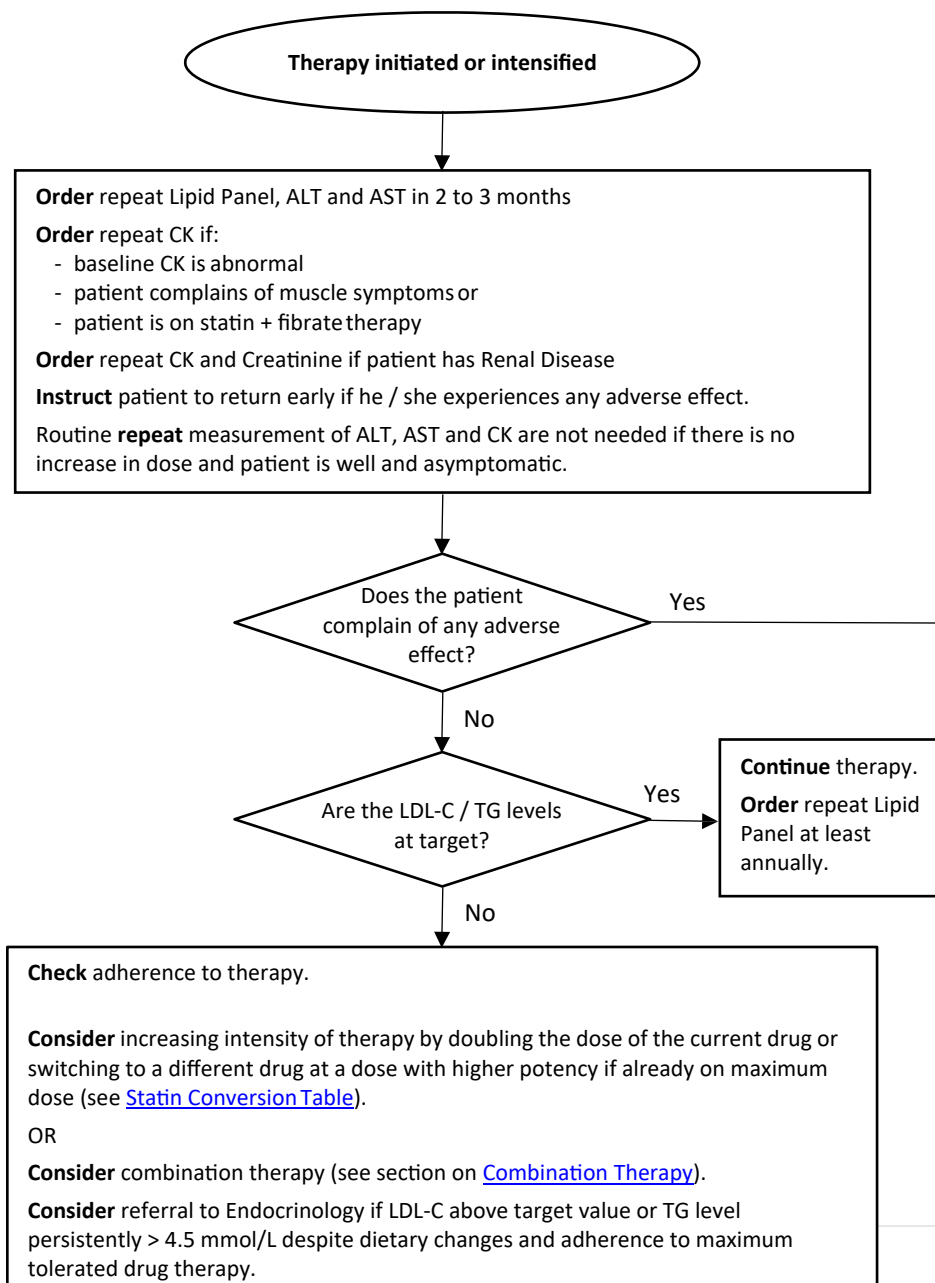
Simvastatin Dose Limitations
Contraindicated with: • Itraconazole • Ketoconazole • Posaconazole • Erythromycin • Clarithromycin * • Telithromycin • HIV protease inhibitors • Nefazodone • Gemfibrozil • Cyclosporine • Danazol
Do not exceed 10 mg simvastatin daily with: • Verapamil • Diltiazem
Do not exceed 20 mg simvastatin daily with: • Amiodarone • Ranolazine • Amlodipine
Do not exceed 40 mg simvastatin daily with: • Ticagrelor
Avoid large quantities of grapefruit juice (> 1 litre daily)

** If treatment with clarithromycin cannot be avoided, statin therapy with lovastatin or simvastatin must be suspended during the course of treatment.*

Initiating Therapy



Monitoring of Side-Effects of Therapy



Abnormal liver enzymes:

Stop therapy if transaminases exceed 3 times upper limit of normal range.

ALT / AST should thereafter be repeated within 2-3 weeks and monitored till the transaminitis resolves to normal level after cessation of medication. Statin therapy may be re-started at lower doses if the levels returned to normal.

Stop statin therapy if serious liver injury with clinical symptoms and/or hyperbilirubinemia or jaundice occurs during treatment. If an alternate aetiology is not found, the statin should not be restarted.

Muscle symptoms or elevated CK:

- If mild to moderate muscle symptoms develop during therapy and/or CK is elevated 3 times above the upper limit of the normal range or 800 IU/L whichever is lower:

- Discontinue therapy.
- Evaluate the patient for other conditions that might increase the risk for muscle symptoms (e.g., hypothyroidism, reduced renal or hepatic function, rheumatologic disorders such as polymyalgia rheumatica, steroid myopathy, vitamin D deficiency, or primary muscle diseases).
- If muscle symptoms resolve with discontinuation of statin therapy, and if no contraindication exists, give the patient the original or a lower dose of the same statin to establish a causal relationship between the muscle symptoms and statin therapy. If a causal relationship exists, discontinue the original statin; once muscle symptoms resolve, use a low dose of a different hydrophilic statin (e.g., Rosuvastatin or Pravastatin).
- Once a low dose of a statin is tolerated, gradually increase the dose as tolerated.
- If, after 2 months without statin treatment, muscle symptoms or elevated CK levels do not resolve completely, consider other causes of muscle symptoms listed above. Consider the possibility of statin-induced autoimmune myositis.
- If persistent muscle symptoms are determined to arise from a condition unrelated to statin therapy, or if the predisposing condition has been treated, resume statin therapy at the original dose.
- If unexplained severe muscle symptoms or fatigue develop during statin therapy, and/or if creatine kinase is more than 10 times upper limit of normal range, promptly

discontinue the statin and refer patient to the Emergency Department.

Others:

Stop statin therapy if patient reports cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These reported symptoms are generally not serious and are reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

Combination Therapy

- Statins lower the risk of major CV events by 20 – 25% per 1 mmol/L reduction in LDL-C. When statin therapy fails to achieve LDL-C target on the maximum tolerated dose, consideration should be given to use other therapies such as an add-on drug to achieve the LDL-C target level for the patient.
- Ezetimibe is an oral, cholesterol absorption inhibitor that lowers LDL-C by 15 – 20%, with associated CV risk reduction when combined with statin therapy. Proprotein convertase subtilisin/kexin type-9 (PCSK9) inhibitors include monoclonal antibodies (evolocumab, alirocumab) and inclisiran. This class of medication reduce LDL-C by an additional 45 – 60% when added to statin therapy and may be considered for select patients with atherosclerotic cardiovascular disease or FH who require more intensive lipid lowering.
- In patients with atherosclerotic cardiovascular disease and low levels of LDL-C < 2.1 mmol/L (~ 80 mg/dL), the HPS2-THRIVE and AIM-HIGH studies showed no incremental clinical benefit from addition of niacin to other LDL cholesterol-lowering therapy (statin or statin-ezetimibe combination).

Special Considerations

Children

Children who have a first degree relative diagnosed with familial hypercholesterolemia can be screened starting from the age of 2 years (see section on [Familial Hypercholesterolemia](#)).

Healthcare professionals should offer all children with **possible**, **probable**, or **definite** FH a referral to a specialist to make a recommendation on the need for therapy and to initiate therapy if required.

Women

Statins are contraindicated in women who are pregnant, planning to conceive, or breastfeeding.

Elderly

In the elderly (age > 75 years), the decision to start treatment should consider the potential risk-reduction associated with treatment, risk of adverse effects, drug-drug interactions, functional status, and patient preferences.

When used, lipid lowering medications in the elderly should be started at the lowest dose and titrated gently to goal LDL-C to minimise statin-associated side effects. For patients on treatment with a statin and LDL-C < 2.1 mmol/L when they turn > 75 years of age, there is no need to reduce therapy if the treatment is well-tolerated without any adverse effects.

Renal Disease

In patients with end stage chronic kidney disease on dialysis, statins did not significantly improve cardiovascular outcomes.

The starting dose of statins in chronic kidney disease should be low. Atorvastatin and simvastatin (max dose 40 mg / day) do not need renal adjustment. During therapy, serum creatine kinase (CK) and renal function should both be carefully monitored.

Fibrates can be used in patients with chronic kidney disease in stages 1 to 3, but the dosages should be reduced, with appropriate monitoring for side effects, especially myopathy. When creatinine clearance is less than 30 ml/min, fibrates are contraindicated.

Liver Disease

In patients with dyslipidaemia and chronic liver disease, if the level of the two transaminases (ALT and AST) is elevated but < 3 times the upper limit of the normal range, statins can be given but the starting dose should be low. Careful monitoring of the serum transaminases and CK after commencement is recommended.

Fibrates can be given in patients whose transaminase levels are elevated < 3 times the upper limit of the normal range, but at a lower starting dosage. Careful monitoring of the serum transaminases and CK after commencement is recommended.

Role of Health Team Members

1. Family Physician

- Provides all aspects of primary medical care from screening, diagnosis and management of dyslipidaemia, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Offer and arrange Teleconsult for suitable patients
- Perform general screening (i.e., fall risk, social economics, smoking & drinking history)
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal) and cancer screenings (FIT kit, mammogram, PAP smear / HPV)

3. Care Manager

- Patient education on dyslipidaemia, LDL / TG target and lifestyle measures
- Alternate CM visit with Doctor for patient with stable hypertension and/or dyslipidaemia.

4. Advanced Practice Nurse

- Manage patients with diabetes, hypertension and dyslipidaemia within the scope of practice
- Initiate and titrate lipid-lowering medications
- Manage simple common acute conditions such as upper respiratory tract infection, gastroenteritis should the patient concomitantly present with these reasons for encounter

5. Clinical Pharmacist

- Manage patients with diabetes, hypertension and dyslipidaemia
- Initiate and titrate lipid-lowering medications
- For patients who need drug optimisation or closer monitoring due to increased risk for drug interactions, polypharmacy, lack of drug-related knowledge or drug non-adherence

6. Dietitian

- Patient education on lipid-lowering diet
- Patient education on weight management

7. Physiotherapist

- Screening for fitness
- Exercise prescription for weight management
- Patient education on weight management

8. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug/drug-disease interactions
- Perform medication reconciliation.

REFERRAL

Referral to Hospital

- Healthcare professionals should offer all people with possible or definite FH a referral to a specialist to offer genetic testing and cascade screening.
- If the patient's pre-treatment ALT / AST are 1.5 to 3 times (or 150 IU/L whichever is lower) above the upper limit of normal (or post-treatment persistently more than 3 times above the upper limit of normal) consider referral to a Gastroenterology department for further evaluation.
- If the patient's pre-treatment CK level is 5 or more times above the upper limit of normal, consider referral patient to General Medicine Department for further evaluation.
- If the patient's CK level is 10 times or higher than normal upper limit, refer patient to Emergency Department to rule out rhabdomyolysis.
- Consider referral to Endocrinology if LDL-C above target value or TG level persistently > 4.5 mmol/L despite dietary changes and adherence to maximum tolerated drug therapy.

RECOMMENDED CARE COMPONENTS FOR LIPID DISORDERS

Recommended Care Components	Minimum Frequency*	Remarks
Lipid Profile~	Annually	All patients should be risk stratified (as recommended in the Lipids CPG) and receive disease and lifestyle education Targets of treatment should be personalised by levels of risk
Smoking Assessment~	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Serum transaminases [#]	Before starting statins and as clinically indicated (e.g. symptoms suggestive of hepatotoxicity, increase in statin dose)	Especially when the statin dose is increased or when combination therapy is initiated Stop the statin / fibrate if patient is symptomatic
Serum creatine kinase [#]	Before starting statins and as indicated (e.g. muscle symptoms)	Look out for rapid increase in creatine kinase post-initiation or increase of statin or fibrate. Stop the medication if the CK is three times ULN) or at about 800 IU/L (whichever is lower)

* More frequently if clinically indicated

~ These are reportable clinical indicators

[#] These are non-reportable clinical indicators

ABBREVIATIONS

ALT	Alanine transaminase
ApoB	Apolipoprotein B
AST	Aspartate transaminase
BP	Blood pressure
CAD	Coronary artery disease
CK	Creatine kinase
CV	Cardiovascular
DHA	Docosahexaenoic acid
eGFR	Estimated glomerular filtration rate
EPA	Eicosapentaenoic acid
FH	Familial Hypercholesterolemia
HDL	High density lipoprotein
HMG-CoA	3-hydroxy-3-methyl-glutaryl-CoA
LDL	Low density lipoprotein
PCSK9	Proprotein convertase subtilisin/kexin type 9
TC	Total cholesterol
TG	Triglyceride

REFERENCES

1. [Ministry of Health Clinical Practice Guidelines on Lipids](#), Dec 2016
2. Ministry of Health Clinical Practice Guidelines on Screening for Cardiovascular Disease and Risk Factors, Mar 2011
3. [Lipid management: Focus on cardiovascular risk](#). 15 December 2023. Agency for Effective Care, MOH
4. [Chronic Disease Management Programme – Handbook for Healthcare Professionals](#). Ministry of Health Singapore 2022
5. National Population Health Survey. Ministry of Health Singapore 2022
6. FDA Drug Safety Communication
7. Statin induced myopathy, BMJ 2008; 337:a2286
8. Jellinger, Paul S., *et al.* "American Association of Clinical Endocrinologists and American College of Endocrinology Guidelines for Management of Dyslipidaemia and Prevention of Cardiovascular Disease." Endocrine Practice 23.s2 (2017):1-87.
9. Drug Information Handbook, 22nd Edition. Lexi-Comp, Hudson, Ohio. 2013